Roll No. .....

MAI–301

M. B. A. (AI & DS) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

STRATEGIC MANAGEMENT

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any one sub questions among (a) & (b) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section—A

1. (a) Strategy is the determinator of the basic long-term goals of an enterprise, and the adoption of courses of action and the allocation of resources necessary for carrying out these goals. Comment. (CO1)

OR

(b) How different levels of strategies are defined in the hierarchy of the organization and what is the relevance of these levels of strategies in modern business scenarios ? (CO3)

P. T. O.

2. (a) What is the significance of strategic intent ? Discuss the various elements of strategic intent. (CO2)

OR

(b) What are essential tasks while formulating strategies and how these strategies can be fixed for the organization ? (CO4)

3. (a) What are the 5 P's of strategy ? Explain its significance for business organizations. (CO1)

OR

(b) Explain the various conditions where emergent strategies can be adopted by the business organizations. (CO2)

Section—B

4. **Case Study :** (20 Marks)

The smartphone industry is one of the most competitive and dynamic sectors in the global technology market. With numerous players, rapid technological advancements, and shifting consumer preferences, understanding the competitive forces at play is essential for companies operating in this space. In such a scenario a company that specialized in wireless devices mainly used for security purposes wants to enter the manufacturing of smartphones. The company took this strategic move to increase its profit by diversifying its segments into consumer markets. The company does not have prior

experience in dealing with customers as they have been in B2B model for the last couple of years. The company is looking for better suggestions so that they can make the new business a big success.

(i) Being a strategic analyst you are required to guide the company in conducting Porter's five forces analysis to evaluate the attractiveness of the smartphone industry. (CO5)

(ii) Do you think that only deliberate strategies are fit for this organization ? Justify your answer in this case. (CO4)

Roll No. .....

MAI–302

M. B. A. (AI & DS) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

PRIVACY ETHICS & REGULATION IN AI

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-questions among (a) & (b) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section—A

1. (a) "All agreements are not contract but all contracts are agreements." Discuss the statement explaining essential elements of a valid contract ? (CO1, 2, 6)

OR

(b) D lived as a paying boarder with a family. He agreed with the members of the family to share prize money of a newspaper competition. The entry sent by D won a prize of ₹ 10,000. He refused to share the amount won. Can the members of the family recover their share ? (CO1, 2, 6)

P. T. O.

2. (a) Explain the nature of a contract of sale of goods and bring out clearly the distinction between a sale and agreement to sell. (CO2, 5, 6)

OR

(b) What are the functions of the Competition Commission of India ? What are its powers ? (CO 2, 5, 6)

3. (a) How is company formed under the Companies Act, 2013 ? Enumerate the various documents to be filed with the Registrar. (CO4, 5, 6)

OR

(b) What is the composition of the National Consumer Disputes Redressal Commission (The National Commission) ? What is its jurisdiction and what procedure does it follows to settle complaint ? (CO4, 5, 6)

Section—B

4. **Case Study :**

(20 Marks) (CO1, 3, 4, 6)

Assume you are working as IT Head at DCodeAI, which is an AI learning app. Due to the ongoing pandemic, all employees are working remotely. Remote employees throughout the company should be able to connect with one another in a smooth and simple manner, without being bombarded with irrelevant or unwanted communications. However, with so many professionals working there, emails can get mixed up and even lost. This leads to loss of productivity plus a couple of communication issues. You propose that all employees use an instant communication app so that only official mails are sent over the email. Taking this as a problem, choose which of the following would be the ideal problem statement template and give valid justification to it.

(i) Remote employees have a problem that they are getting irrelevant and unwanted mails frequently from different departments. An ideal solution is to block everyone.

(ii) Remote employees have a problem that they are getting irrelevant and unwanted mails frequently from different departments. An ideal solution is to fire some professionals.

(iii) Remote employees have a problem that they are getting irrelevant and unwanted mails frequently from different departments. An ideal solution is to use the communication app and email should be used only for formal work messages.

(iv) Remote employees have a problem that they are getting irrelevant and unwanted mails frequently from different departments. An ideal solution is to write letters and post them in the letterbox.

Roll No. .....

MAI–303-AI(1)

M. B. A. (AI & DS) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

DATA VISUALIZATION FOR MANAGERS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-question among (a) & (b) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 marks.

Section—A

1. (a) Can you describe a scenario where data visualizations helped us solve a specific business challenge ? (CO3)

OR

(b) Explain the basic architecture of a neural network used in deep learning. (CO1)

2. (a) Can you provide an example of a successful data visualization that improved decision-making in our organization ? (CO1/CO2)

P. T. O.

OR

(b) What is the purpose of activation functions in neural networks, and can you name a few commonly used ones ? (CO1)

3. (a) Write the python code to perform the calculation of F (x,y,z) = (x+y) *z using Tensor Flow. (CO3)

OR

(b) What are PyTorch tensors, and how do they relate to NumPy arrays ? (CO2, CO3)

Section—B

(20 Marks)

4. **Case Study :** (CO1/CO3)

Case Study : Predicting crop yields for apples and oranges.

Problem Statement : Create a linear regression model that predicts crop yields for apples and oranges (target variables) by looking at the average temperature, rainfall, and humidity (input variables or features) in a region.

Here's the training data :

Region	Temp. (F)	Rainfall (mm)	Humidity (%)	Apples (ton)	Oranges (ton)
Kanto	73	67	43	56	70
Johto	91	88	64	81	101
Hoenn	87	134	58	119	133
Sinnoh	102	43	37	22	37
Unova	69	96	70	103	119

(a) Write the yielding equations for predicting crop yields for apples and oranges.

(b) What will be the loss function in this case ?

(c) How gradient or derivative of the losses with respect to the weights and biases will help to minimizes the errors ?

Roll No. .....

MAI–304-AI-2

M. B. A. (AI & DS) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

BIG DATA ANALYTICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-questions among (a) & (b) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section—A

1. (a) Discuss the concept of "Big Data" and explain how it differs from traditional data analytics. Provide examples of the types of data that would be considered "Big Data" and explain how its size, complexity, and variability pose challenges to analysis. (CO1,2)

P. T. O.

OR

(b) Describe the various stages of the Big Data analytics process, including data ingestion, data cleaning and preprocessing, data analysis, and data visualization. For each stage, explain the challenges that must be overcome to ensure accurate and meaningful results. (CO1, 2)

2. (a) Explain the differences between supervised and unsupervised learning algorithms in the context of Big Data analytics. Provide examples of each type of algorithm and explain how they can be used to extract insights from large datasets. (CO1, 3)

OR

(b) Describe the concept of "data mining" and explain how it can be used to uncover patterns, relationships, and anomalies in large datasets. Provide examples of techniques used in data mining and discuss their strengths and weaknesses. (CO1, 3)

3. (a) Discuss the role of data visualization in Big Data analytics. Explain how visualizations can be used to help analysts identify trends, patterns, and outliers in large datasets. Provide examples of different types of visualizations and describe when they are most appropriate to use. (CO1, 3)

OR

(b) Explain the concept of "predictive analytics" and describe how it can be used to forecast future trends and events. Provide examples of industries that use predictive analytics and discuss the challenges associated with creating accurate predictions from large datasets. (CO1, 3)

Section—B

4. **Case Study** : Big Data Analytics using Hadoop in Retail Industry.

Company Overview

The retail company is a global brand with stores in multiple countries. The company has a large customer base and sells a wide range of products, including electronics, apparel, and groceries. The company's online presence is also significant, with a growing number of customers choosing to shop online.

Business Problem

The company wanted to gain insights into customer behavior, preferences, and purchasing patterns to optimize its marketing strategies and improve customer experience. However, the company's existing data infrastructure was unable to handle the large volume of data generated from various sources such as in-store sales, online transactions, social media, and loyalty programs. The company needed a scalable, cost-effective, and efficient solution to manage and analyze this data. Suggest suitable solution for this problem.

(CO1, 3)

Question :

What is Big Data ? Explain its characteristics and types.

Roll No. .....

MAI–305(F1)

M. B. A. (AI & DS) (THIRD SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2023

FINANCIAL ANALYTICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-questions among (a) & (b) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section—A

1. (a) What are different types of data used in Financial Analytics ? (CO1)

OR

(b) Describe Financial Analytics Process in specific business scenario considering relevant finance and economic data. (CO2)

2. (a) Describe the utility of Financial and Accounting Management in Decision Making process in Business. (CO1)

P. T. O.

OR

(b) As a Bank Manager, how would you decide whether to give Credit to a company or not and what steps you would follow to identify a fraud for a company ? (CO2)

3. (a) Describe the advantage and limitations of using Finance Data in Decision Making process. (CO2)

OR

(b) As a Finance Manager, how would you decide whether to give Discount to a Customer or not and what steps you would follow to identify a customer for a company. (CO3)

Section—B

4. **CASE : FINANCE ANALYTICS** (CO2/CO3)

You are working for a Retail Company. Overall sales have been declining for the last three quarters and management is very much concerned. Each retail store has its own individual databases that keep track of sales for specific items. However, at the overall management level, only sales figures for each store are reported. Management has asked you to prepare reports by sale items (such as specific brand X and model Y of running shoes) and by categories of sale items (such as all running shoes) so that they can make accurate decisions on which product lines to drop to reduce overall inventory costs. As a Analyst or as a part of a development team, you will need to develop a procedure to extract data from each individual database, reconcile data formats and types, aggregate all data into a data repository, and develop queries based on management requests. You will communicate and work closely with management representatives to ensure that the created

data repository and reports meet their needs. Depending on your level of responsibilities, you will develop or contribute to the development of a realistic estimate of time, resources and cost to achieve the intended goal. Once the project is launched, you will develop full specifications, develop and test subsystems, and integrate into final product. Throughout the development process, you will conduct a series of increasingly rigorous technical, functional and user tests. Based on the results of these tests, you can expect revision of technical specifications, and overall system and unit design, along with project timeline, resources and cost revisions. Of particular importance will be a focus on user needs and requirements, business applications, and easy-to-use user interface. Your BI team may be also invoked in developing and delivering training to managers and other stakeholders in using the new system.

(i) Explain the procedure for extracting data from each individual database as Analyst.

(ii) What role and responsibility does analyst plays in Management of Data ? Justify.

(iii) What all factors will lead to reduction of the stock and other relevant financial cost ?

Roll No. .....

MAI–306(F2)

M. B. A. (AI & DS) (THIRD SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2023

FINANCIAL PLANNING

Time : 1½ Hours

Maximum Marks : 50

Note : (i) This question paper contains two Sections–Section A and Section B.

(ii) Both Sections are compulsory.

(iii) Answer any *one* sub-questions among (a) & (b) in each main question of Section A. Each question carries 10 marks.

(iv) Section B consisting of case study is compulsory. Section B is of 20 Marks.

Section—A

1. (a) Elaborate on the primary objectives of financial planning and why they are essential for individuals and families. Support your explanation with real-world examples illustrating the significance of financial planning.

(CO1)

P. T. O.

OR

(b) In a detailed manner, describe the multifaceted responsibilities of a financial planner. Explore how a financial planner's role evolves when dealing with clients in different financial situations or life stages. Provide examples to elucidate your points. (CO2/CO3)

2. (a) Examine the various stages of the consumer life cycle and delve into the specific financial needs and priorities associated with each stage. Illustrate your analysis with comprehensive examples to demonstrate the evolving financial challenges individuals face as they progress through different life stages. (CO3/CO4)

OR

(b) Thoroughly outline each crucial step in the financial planning process, discussing the purpose and significance of each step. Elaborate on the establishment of a client-planner relationship and its impact on the overall success of financial planning. (CO3/CO4)

3. (a) Provide a comprehensive explanation of the various types of client objectives and needs that must be considered during financial planning. (CO2/CO3)

OR

(b) Explain the role of risk assessment in the financial planning process. Discuss various types of financial risks that clients may face and the methods financial planners use to assess and mitigate these risks. (CO2/CO3)

Section—B

4. **Case Study :**

Financial Planning for a Mid-Aged Software Engineer

Background :

Rajesh Sharma is a 45-year-old software engineer working for a prominent IT company in Bengaluru, India. He has been in the industry for over 20 years, steadily climbing the corporate ladder. Rajesh is married to Meena, a homemaker, and they have two teenage children, Ravi and Ayesha. While he has had a successful career, Rajesh is becoming increasingly concerned about his financial future and wants to ensure a comfortable retirement. He also wants to secure his children's higher education and maintain a good quality of life for his family.

Financial Situation :

Annual Income : INR 25 lakhs (including bonuses and benefits)

Savings and Investments : INR 50 lakhs (diversified in mutual funds, fixed deposits, and a few stocks)

Outstanding Home Loan : INR 30 lakhs

Monthly Household Expenses : INR 60,000

No significant health issues; has a standard employer-provided health insurance plan

Financial Goals :

(i) Retirement Planning : Rajesh desires to retire at the age of 60 with a comfortable nest egg that allows him and Meena to maintain their current lifestyle and travel in their retirement years.

(ii) Children's Education : He aims to save for Ravi and Ayesha's higher education in top institutions, including funding for study abroad if necessary.

(iii) Home Loan : Rajesh wishes to clear his home loan before retirement to reduce financial burden in his post-retirement years.

Scenario : Rajesh has recently consulted a certified financial planner and that is you to help him create a comprehensive financial plan that addresses his goals and concerns. You have assessed Rajesh's current financial situation and is ready to provide recommendations for a personalized financial plan.

(a) Analyze the financial planning strategies that Rajesh should adopt to ensure a secure and comfortable retirement. Consider factors like inflation, life expectancy, and investment options. Provide recommedations on how he can reach his retirement goals by age 60.

(CO5/CO6)

(b) Examine and evaluate the various investment and savings options available to Rajesh for finding Ravi and Ayesha's higher education, which includes potential international education costs. Discuss the benefits and risks associated with these options, and recommend a customized plan that aligns with his children's education goals.

(CO5/CO6)